HL-SAM-Seg: Complementary High- and Low-Resolution Features Based on SAM for Remote Sensing Image Semantic Segmentation

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***Abstract*—The groundbreaking segment anything model (SAM), built on a vision transformer (ViT) design with millions of parameters and trained on the large SA-1B dataset, acts as a vision foundation model that can be used for various segmentation tasks. However, this model cannot be directly applied to the semantic segmentation of remote sensing images, as it tends to oversegment objects rather than preserving their semantics. Furthermore, it does not align well with object boundaries, which are usually required for the high-accuracy segmentation of high-resolution remote sensing images. To address this issue, we adjust the image encoder and mask decoder of SAM and propose a high-resolution and low-resolution SAM-based semantic segmentation (HL-SAM-Seg) network. The image encoder is extensively adjusted to comprise an adapter, a high-resolution (high-res) path, and a low-resolution (low-res) path. The latter two paths extract and update the high-res and low-res features, which are merged in the mask decoder to perform multiclass segmentation. Specifically, we design a high-resolution and low-resolution (HL) module inserted into the transformer blocks of the low-res path to align and update the low-res and high-res features. Experimental results on the International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen, ISPRS Potsdam, and FloodNet datasets show that the proposed HL-SAM-Seg outperformed conventional state-of-the-art semantic segmentation algorithms overall, with underperformance in some small sample categories. Moreover, it has fewer trainable parameters, suggesting the potential for leveraging SAM for remote sensing image segmentation.**

***Index Terms*—Deep learning, remote sensing image, segment anything model (SAM), semantic segmentation.**

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I. Introduction

SEMANTIC segmentation is a fundamental computer vision task that involves assigning a semantic class label to every pixel in an image, focusing on pixel-level understanding and spatial coherence [1], [2]. The output is a dense pixel-wise prediction map where all pixels belonging to the same semantic class are grouped together. In the remote sensing community, semantic segmentation is the technical process that is usually applied to achieve pixel-level land cover/land use classification, producing a thematic map from an aerial or satellite image [3], [4]. In the contemporary literature [5], [6], [7], [8], the task of generating a high-resolution land cover map is formulated as a remote sensing image semantic segmentation problem, which performs pixel-wise classification while leveraging spatial–contextual features. In recent years, remote sensing image semantic segmentation has advanced rapidly with the development of numerous deep learning networks [9], [10], [11], [12]. It has been widely used for land cover classification [3], [4] and change detection [13], [14], [15], supporting a variety of applications including national defense, land surveying, disaster monitoring, and transportation planning [16], [17], [18], [19].

The proliferation of extensive image datasets has led to the emergence of vision foundation models, such as the segment anything model (SAM) [20], that can comprehend and engage with the visual realm in unprecedented ways. Applying SAM to remote sensing image segmentation is an opportunity as well as a challenge [21]. SAM comprises a ViT-based [22] image encoder, a lightweight mask decoder, and a versatile prompt encoder to generate masks from diverse inputs, including points, boxes, masks, and text. The image encoder of SAM can extract rich semantic features due to its training on billions of images. Utilizing the SAM backbone eliminates the necessity of acquiring extensive remote sensing datasets to train such a substantial model [23], [24], [25], [26].

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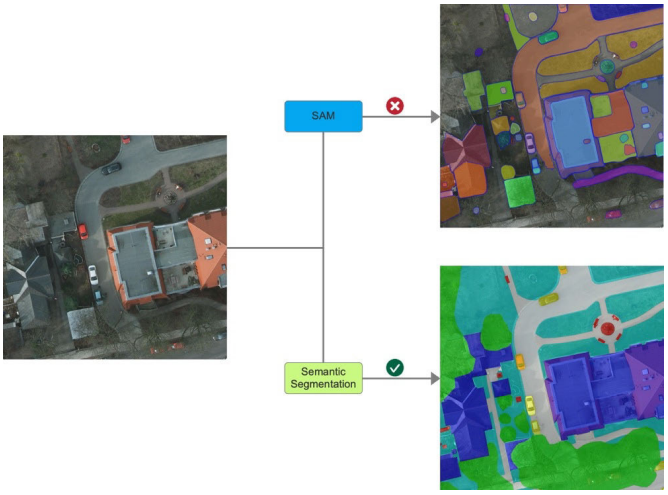


Fig. 1. Segmentation results of the "segment everything" of SAM and the semantic segmentation GT of the remote sensing image. Compared to the GT, SAM is unable to classify ground objects and tends to oversegment the semantic objects, such as the roof in the lower left corner.

To adapt SAM to specific downstream tasks while mitigating the computational burden of full model fine-tuning, researchers have explored techniques such as adapter modules [23] and soft prompts [24]. For the adapter modules approach, the model can be adapted to different downstream tasks by inserting trainable adapter modules into the frozen backbone. The soft prompts approach involves training the model's prompt encoder to generate task-relevant prompts. Soft prompts adjust the model depending on the input prompts, and it is more suitable for simple tasks, such as image classification, rather than dense prediction tasks, such as image segmentation. Therefore, the adapter modules are more suitable for applying in semantic segmentation based on SAM.

In addition, the image encoder is based on the ViT. With the self-attention module, ViT achieves direct global relationship modeling; hence, it is good at capturing contextual information [27], [28], [29]. However, the ViT backbone primarily focuses on learning global dependencies via channel-wise attention, which is insufficient for capturing fine-grained spatial details [30], [31], such as object boundaries. In contrast to ViT architecture, which is superior in capturing global information and needs a large computational cost and a large-scale training dataset, the convolutional neural network (CNN) architecture excels at capturing local features, and it performs well with a small-scale dataset [32], [33], [34]. These complementary features have driven the integration of CNNs and transformers, such as Unetformer [5], CNN and multiscale Transformer fusion network (CMTFNet) [6], and a full-scale connected CNN–Transformer network (FSC-CNN) [35]. This combination is particularly suitable for remote sensing image segmentation, which needs to balance between the accuracy of semantic segmentation and the precision of the segmented boundaries. However, these models require retraining for different datasets, without taking advantage of the existing vision foundation models.

The mask decoder of SAM performs foreground segmentation based on a given prompt, frequently leading to the oversegmentation of semantic objects (as illustrated in Fig. 1). Although there are previous studies about training the prompt encoder and mask decoder of SAM to generate remote sensing-related prompts [24], [25], they lack improvements in semantic segmentation tasks, such as addressing the mismatch between contours and semantics. For remote sensing image segmentation, understanding the contextual information is required to achieve accurate semantic segmentation and preserve fine-grained details to guarantee the precision of pixel segmentation. Therefore, the primitive SAM image encoder and decoder need to be modified to support remote sensing semantic segmentation.

In this study, we propose an HL-SAM-Seg network to improve SAM with mechanisms that effectively preserve fine-grained details in the remote sensing image segmentation. First, to incorporate the SAM's generalization ability based on its large training dataset, an adapter path composed of several adapters is added in the image encoder to adjust the encoded features. Second, we add a high-resolution (high-res) path that is composed of several lightweight one-convolutional layers (OCLs) to better preserve the image details. Particularly, we design a high-resolution and low-resolution (HL) module to achieve mutual updating between the high-res and low-resolution (low-res) features extracted from the backbone of SAM. Finally, we modify the mask decoder to receive both high-res and updated low-res features, enabling multicategory mask output. The proposed method is tested with the publicly available International Society for Photogrammetry and Remote Sensing (ISPRS) Vaihingen and Potsdam datasets. In addition, we also test the proposed method with the FloodNet dataset [36]. To validate the performance of the proposed network, it is compared against other state-of-the-art networks, including DeepLabv3+ [1], U-Net [2], UnetFormer [5], CMTFNet [6], boundary-enhanced dual-stream network (BEDSN) [7], and RS3Mamba [8], through experiments with these two datasets.

The remainder of this article is organized as follows. Section II describes the proposed HL-SAM-Seg network in detail. Section III briefs the experiment setup. Then, the experiments and comparison between different networks are given in Section IV. Section V presents an ablation study of each module. In Section VI, the limitations of the proposed method are discussed. Finally, the conclusions are summarized in Section VII.

II. Methods

In this section, the overall structure of the HL-SAM-Seg will be presented first, and then, its core modules, namely, the adapter, the OCL module, the HL module, the mask decoder, and the loss function, will be described in detail.

A. Overall Structure

The overall network structure of HL-SAM-Seg is shown in Fig. 2. It primarily consists of an image encoder and a mask decoder. The image encoder contains an adapter path, a low-res path, and a high-res path. The adapter path is employed to generate task-specific features. In this path, the input image undergoes a linear module to generate embedding features.

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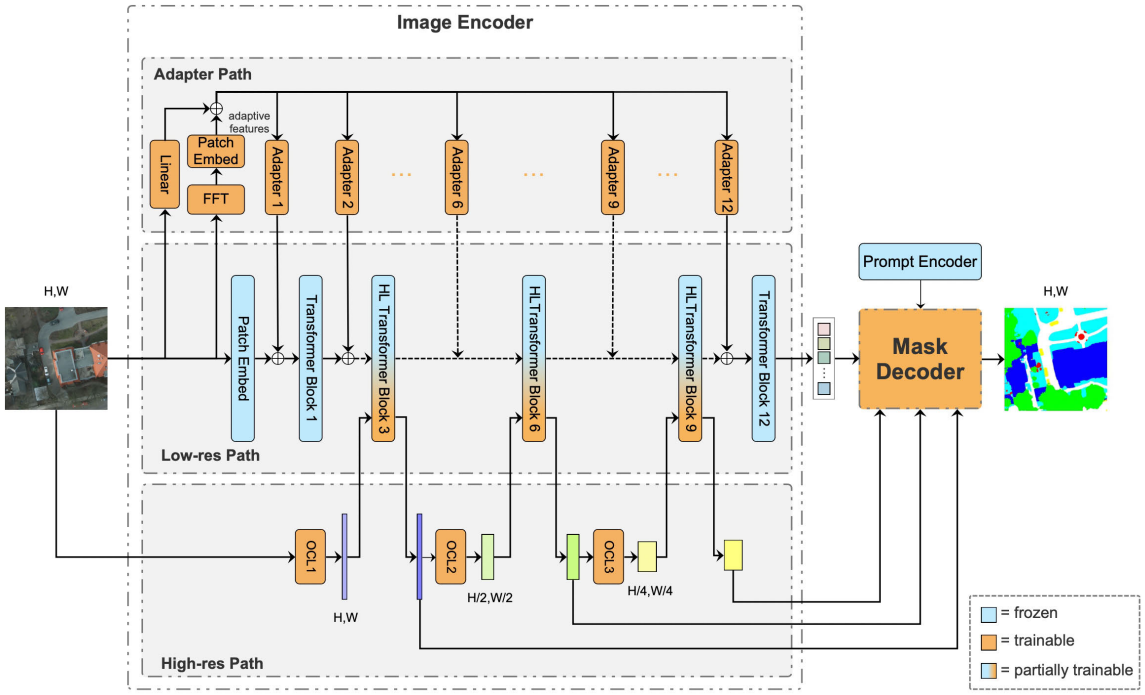


Fig. 2. Architecture of the proposed HL-SAM-Seg model.

Simultaneously, a composition of fast Fourier transform (FFT) and patch embedding is used to generate handcrafted features. The sum of the embedding features and the handcrafted features serves as the adaptive features, which are then input to the different adapters to generate task-specific features, which are then inserted into different layers in the low-res path. The low-res path utilizes the frozen pretrained image encoder of SAM to extract the 16× downscaled features as low-res features. The high-res path, composed of OCL modules, extracts high-res features. Additionally, in the low-res path, we design an HL module and insert it in the third, sixth, and ninth transformer blocks to update the low-res and high-res features, avoiding the loss of details. The HL module will be introduced in detail in Section II-D.

Finally, the updated low-res and high-res features that are output by their respective paths are sent to the mask decoder. In the mask decoder, the low-res features are processed in the same way as the SAM. Differently, we replace the single MLP in SAM's mask decoder, which outputs foreground probabilities, with multiple MLPs, one for each class, to output low-res predicted masks for each category. Meanwhile, to fuse semantic and detailed information more thoroughly, we upsampled the low-res features to merge with the high-res features layer-wise. Subsequently, we passed the merged features through the same set of multiple MLPs for category prediction, outputting high-res prediction results. The low-res prediction results are unsampled and added to the high-res prediction results to obtain the final prediction.

B. Adapter

To adapt the extracted features from the original backbone to the semantic segmentation, we add an adapter path to the image encoder, which first extracts embedding features by a linear module and handcrafted features by a composition of FFT and patch embedding. The embedding features and handcrafted features are then added to form the adaptive features, which are input into adapters $1 \sim 12$ to obtain adapted features to be sent to each transformer block in the low-res path. All adapters share a unified structure: a dimensionality linear reduction layer (Down), a GELU activation function [37], and a shared dimensionality restoration linear layer (Up), as shown in Fig. 3(a). The adapter can be formally expressed as follows:

$$\text{Adapter}\left(f_{\text{input}}\right) = \text{Up}\left(\text{GELU}\left(\text{Down}\left(f_{\text{input}}\right)\right)\right) \tag{1}$$

where the dimensionality reduction linear layer Down is used to generate task-specific prompts for each adapter and Up is a dimensionality restoration linear layer shared across all adapters that adjusts the dimensions of transformer features.

C. OCL Module

The OCL module is designed to extract fine details and edges of the semantic objects. Each OCL module comprises one 2-D convolutional layer, one normalization layer, and one LeakyReLU activation, as expressed in the following equation:

$$\text{OCL}\left(f\right) = \text{LeakyReLU}\left(\text{LN}\left(\text{Conv}\left(f\right)\right)\right). \tag{2}$$

The high-res path consists of three OCL modules. The convolutional layer in the first OCL module employs a 3×3 kernel with a stride and padding of 1, processing an input of three channels to produce 16 output channels. The second and third OCL modules perform downsampling on the input feature maps. The second OCL has 16 input and 36 output channels, while the third has 32 input and 64 output channels. Both modules retain the same kernel size and padding as previous layers but use a stride of 2.

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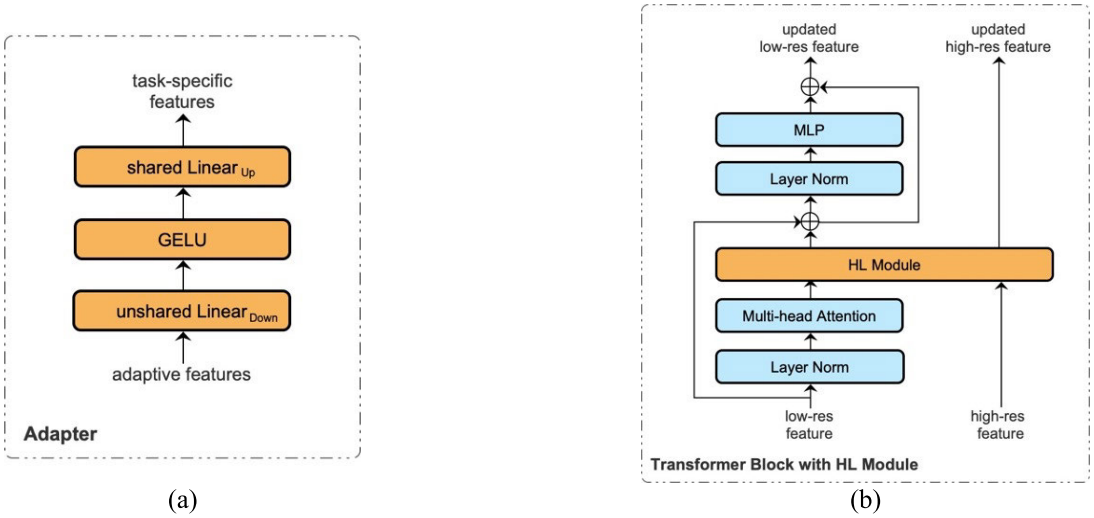


Fig. 3. Structure of (a) adapter and (b) HL module.

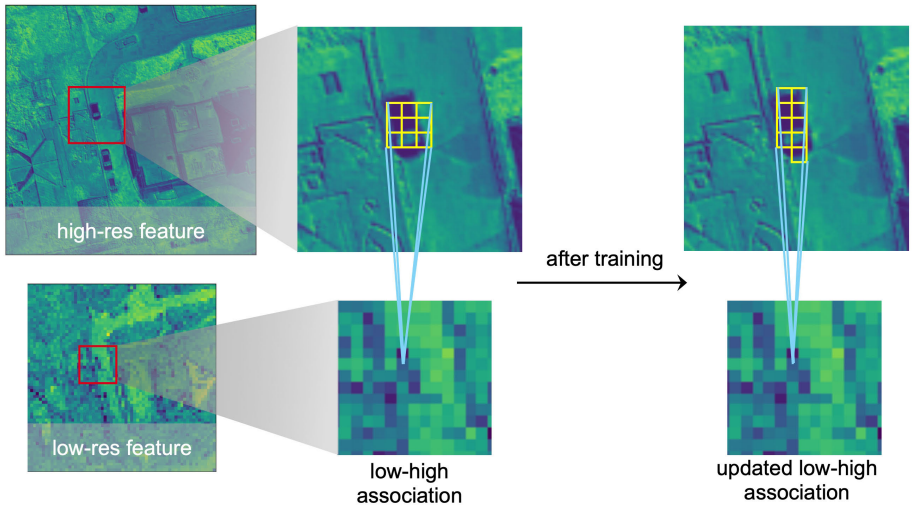


Fig. 4. Example of the low-high association. (Top row) High-res feature map. (Bottom row) Low-res feature map. Initially, one low-res pixel corresponds to the 3×3 neighboring pixels in the high-res feature map. After training, the low-res pixel can be associated with the contour in the high-res feature map.

The input RGB image is first processed by the initial OCL module, resulting in a high-resolution feature map of size $16\times$ H $\times$ W. The first HL module then refines this output. The refined features are passed sequentially to the next OCL module for further processing. This OCL–HL sequence repeats three times along the high-resolution path. After each pass through an OCL and an HL module, the spatial dimensions of the feature map are reduced, resulting in updated sizes of H $\times$ W, H/2 $\times$ W/2, and H/4 $\times$ W/4.

D. HL Module

Effective semantic segmentation requires not only accurate semantic labeling but also precise pixel-level localization. Semantic information is typically captured in high-level, low-resolution features, while pixel-level details, such as contours and boundaries, are preserved in low-level, high-resolution features. In superpixel segmentation, an association map is often used to establish relationships between pixels and superpixels [38], [39], effectively bridging detailed and object-level information. Inspired by this idea, we introduce a cross-attention mechanism [40], [41] to connect low- and high-resolution features by iteratively updating an association map within an HL module. This HL module is integrated into a transformer block, forming what we refer to as the HL-transformer block, as illustrated in Fig. 3(b).

1) Low-Res Features: For each pixel in the low-res feature, we consider it to be associated with its surrounding pixels in the high-res features. The low-to-high association matrix represents the probability of association between the low-res pixel and its corresponding high-res neighborhood. This association is initially established using the 3×3 neighborhood pixels and updated in the training for better assignment, as shown in Fig. 4. The low-res and high-res feature association can be expressed in the following equation:

$$A_{ip} = \mathbf{q}_{L_i} \cdot \mathbf{k}_{H_p \in \mathcal{N}(i)} \tag{3}$$

where A_{ip} denotes the low-to-high association map, which is equal to the dot multiply of $\mathbf{q}_{L_i}$ and $\mathbf{k}_{H_p \in \mathcal{N}(i)}$ that represent the low-res query vector and the key vectors of the high-res neighborhood, respectively. $\mathcal{N}(i)$ denotes the high-res

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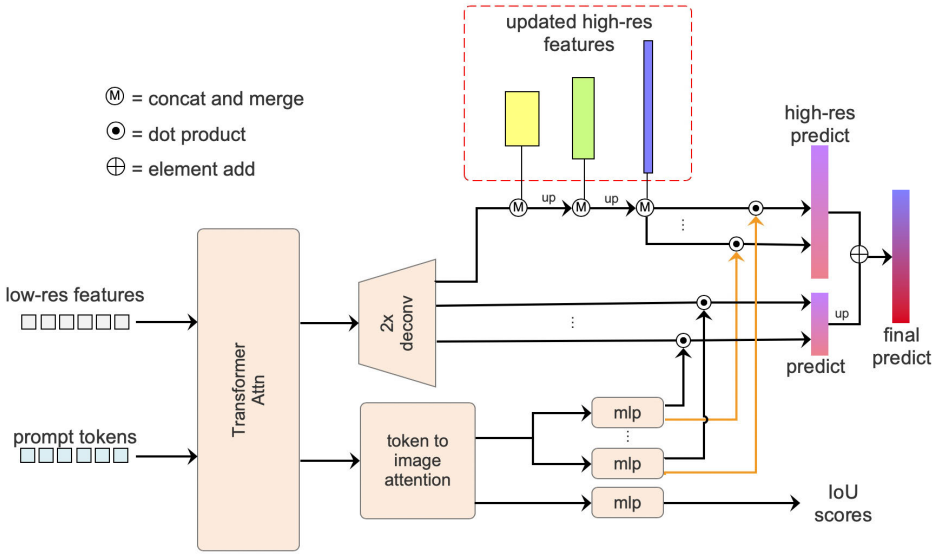


Fig. 5. Mask decoder of the HL-SAM-Seg model adapted from the mask decoder of SAM.

neighborhood pixels surrounding the pixel i in the low-res feature. Then, the transformation from high-res features to low-res features can be calculated by using this association map as in the following equation:

$$\delta L_i = \sum_{p \in \mathcal{N}(i)} A_{ip} \cdot \mathbf{v}_{H_p} \tag{4}$$

where A_{ip} denotes the association map between the pixel i in the low-res feature and the pixel p in the high-res neighborhood and $\mathbf{v}_{H_p}$ represents the high-res value vector. The calculated δL_i can perceive the high-res contour, thus enabling better contour perception. Finally, the original low-res features can be updated by δL_i. By substituting (3) into (4), the complete formula is shown in the following equation:

$$L_i' = L_i + \sum_{p \in \mathcal{N}(i)} \text{softmax}_p \left(\mathbf{q}_{L_i} \cdot \mathbf{k}_{H_p}\right) \mathbf{v}_{H_p}. \tag{5}$$

2) High-Res Features: Similarly, for each high-res pixel, a corresponding set of pixels exists in the low-res neighborhood. The high-to-low association matrix represents the probability of association between the high-res pixel and its corresponding low-res neighborhood. This fine-tunes the high-res features to better align semantic expressions with the high-res at a finer granularity. This association is expressed in the following equation:

$$H_p' = H_p + \sum_{i \in \mathcal{N}(p)} \text{softmax}_i \left(\mathbf{q}_{H_p} \cdot \mathbf{k}_{L_i}\right) \mathbf{v}_{L_i} \tag{6}$$

where H_p' represents the updated high-res features, H_p is the original high-res features, $\mathbf{q}_{H_p}$ represents the high-res query vector, $\mathbf{k}_{L_i}$ denotes the key vectors of the low-res neighborhood, and $\mathbf{v}_{L_i}$ represents the low-res value vector. To save computational resources and improve efficiency, we do not insert the HL module into all transformer blocks. Instead, we insert HL modules into the third, sixth, and ninth transformer blocks to align and update the low-res features processed by global attention with the high-res features of three different scales from the OCL modules.

E. Mask Decoder

To represent multiple classes for each pixel, a multiclass mask decoder is constructed based on the original mask decoder of SAM. This is achieved by using multiple MLPs to perform class prediction for the land cover classification task. In the proposed HL-SAM-Seg algorithm's per-class mask decoder, the IoU score is ignored, resulting in only one mask output that participates in the calculation of training loss. The structure of the improved mask decoder is shown in Fig. 5. Transformer Attn is used for bidirectional perception between low-res features and prompt tokens. Token-to-image attention is used to further enhance the expression of prompt tokens, and deconv is used to upsample low-res features. These are all parts of the original architecture. To output segmentation masks for multiple classes, we replace the original three MLPs that output foreground probabilities after the token-to-image attention with multiple MLPs, one for each class, to output class tokens. These class tokens are then dot-producted with the deconvoluted low-res features to obtain prediction results and dot-producted with high-resolution features fused from multiple scales to obtain high-resolution prediction results. Finally, the prediction results are unsampled and added to the high-resolution prediction results to obtain the final prediction.

F. Loss Function

In this study, we use cross entropy to calculate the loss function [5], [42]

$$\text{Loss} = -\frac{1}{N} \sum_{n=1}^{N} \sum_{k=1}^{K} y_k^n \log \left(p_k^n\right) \tag{7}$$

where N represents the total number of pixels in the batch, K represents the number of classes, y_k^n represents the one-hot encoding of the true labels, and p_k^n denotes the predicted probability that the sample pixel n belongs to class k.

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III. Experiments

A. Experimental Datasets and Settings

In this study, we use the publicly available ISPRS Vaihingen and Potsdam datasets to test the performance of the proposed network. The ISPRS Vaihingen and Potsdam datasets are airborne datasets for the urban land cover classification competition. The challenge in this competition stems from the highly variable appearance of objects such as buildings, streets, trees, and cars in high-resolution data. This results in large intraclass variance and low interclass variance. Vaihingen is a relatively small village characterized by many individual buildings and small multistory structures. Potsdam is a typical historical city with large building complexes, narrow streets, and dense settlement patterns. Each dataset has been manually classified into six land cover classes, including impervious surfaces, buildings, low vegetation, trees, cars, and clutter.

These two datasets are independently tested on the proposed network. The ISPRS Vaihingen dataset comprises 33 orthophotos with an average size of 2500 × 2000 pixels. In our experiments, the dataset was divided into a training set of 16 photographs and a test set of 17 photographs. The ISPRS Potsdam dataset comprises 38 orthophotos with an average size of 6000 × 6000 pixels. We utilized 18 designated images for training, while the remaining 20 served as the test set.

To further evaluate the generalization capability of our proposed network across diverse disaster scenarios, we additionally employ the FloodNet dataset [36], a benchmark dataset specifically designed for post-flood assessment using uncrewed aerial vehicle (UAV) imagery. Unlike the ISPRS datasets, which focus on urban scenarios, FloodNet addresses the critical need for accurate damage assessment in natural disaster scenarios, particularly in flooded areas. The FloodNet dataset comprises high-resolution UAV images captured in the aftermath of Hurricane Harvey in 2017, covering multiple flood-affected regions in Texas, USA. The images exhibit a spatial resolution of 10 cm per pixel, enabling detailed visual analysis for fine-grained semantic segmentation tasks. Each image in the dataset has dimensions of 4000 × 3000 pixels, ensuring sufficient spatial coverage and granularity for robust model training and evaluation. The dataset is annotated with eight semantically distinct classes: building flooded, road flooded, water, vegetation, building nonflooded, road nonflooded, vehicle, and background. These annotations facilitate a comprehensive assessment of flood impacts on both built infrastructure and natural environments.

B. Evaluation Metrics

To evaluate our proposed model, we used three evaluation metrics: overall accuracy (OA), mean $F1$ score (m$F1$), and mean intersection over union (mIoU) to compare it with currently popular models. OA and m$F1$ evaluate the pixel-level classification performance of the models, while mIoU assesses models from the perspective of segmentation quality. These criteria are commonly used in the literature [5], [6], [7], [8], in which mIoU serves as the primary ranking metric, while OA and m$F1$ are also reported for reference. Based on the multiclass confusion matrix, OA, m$F1$, and mIoU are calculated as (8), (11), and (12) [7]. It should be noted that OA represents the proportion of correctly classified pixels out of the total number of pixels [43], [44]. For multiclass segmentation, it is the summation of diagonal elements (TP_k) of the confusion matrix divided by the total number of sample pixels as follows:

$$\mathrm{OA} = \frac{\sum_{k=1}^{K} \mathrm{TP}_k}{H * W} \tag{8}$$

$$\mathrm{precision}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k} \tag{9}$$

$$\mathrm{recall}_k = \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FN}_k} \tag{10}$$

$$\mathrm{m}F1 = \frac{1}{K} \sum_{k=1}^{K} \frac{2 \times \mathrm{precision}_k \times \mathrm{recall}_k}{\mathrm{precision}_k + \mathrm{recall}_k} \tag{11}$$

$$\mathrm{mIoU} = \frac{1}{K} \sum_{k=1}^{K} \frac{\mathrm{TP}_k}{\mathrm{TP}_k + \mathrm{FP}_k + \mathrm{FN}_k} \tag{12}$$

where H and W represent the height and width of the output result, respectively; K denotes the number of classes; and TP_k, FP_k, TN_k, and FN_k denote true positives, false positives, true negatives, and false negatives for samples indexed as class k.

C. Implementation Details

The input image to the proposed model is 1024 × 1024, so we applied a sliding window to crop the images. The cropped images of the ISPRS Vaihingen dataset are divided into a training set of 102 images and a testing set of 108 images. The corresponding numbers for the ISPRS Potsdam dataset are 648 and 720. For the FloodNet dataset, the corresponding numbers are 17 991 and 5610. Image augmentation with mosaic [45] and normalization is performed during training to enhance dataset diversity and improve the model's generalization ability. The mosaic operation involves randomly cropping and stitching four images together to form a new image. Image augmentation can improve the model's generalization ability. In the testing, we do not apply any data augmentation.

SAM provides three versions of pretrained ViT: ViT-B (Base), ViT-L (Large), and ViT-H (Huge), with parameter sizes of 91, 308, and 636 M, respectively. To make the network lightweight, we prefer to use the ViT-B due to its smaller parameter size. Since the SAM model has been trained with a huge volume of images, we leverage the low-res feature extraction on its pretrained model. The prompt encoder is kept according to the original architecture of SAM, without any modification, and does not participate in training. To this end, the weights of the SAM image encoder and prompt encoder are frozen in the training process, while we keep the adapters, OCL modules, HL modules, and the entire mask decoder trainable.

All experiments in this article were trained and validated on a single NVIDIA GeForce RTX 3090 GPU within the PyTorch deep learning framework. Our model was optimized using AdamW with an initial learning rate of 0.0002 and trained for 80 epochs across all datasets, and the batch size is set to 1.

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TABLE I

COMPARISON OF RESULTS OF DIFFERENT METHODS ON THE VAIHINGEN DATASET. IN EACH CATEGORY, THE DIGITS ARE $F1$/MIOU SCORES. THE BEST VALUES IN THE COLUMNS ARE IN BOLD (UNITS: %)

Methods	Impervious surface	Building	Low vegetation	Tree	Car	mF1	OA	mIoU
DeepLabv3+ [1]	89.90/81.66	93.03/86.97	79.52/66.00	85.44/74.58	78.39/64.46	85.26	87.34	74.73
U-Net [2]	88.16/78.84	90.36/82.43	77.89/63.78	84.66/73.41	74.59/59.48	83.13	85.62	71.59
UnetFormer [5]	89.86/81.58	93.00/86.92	79.76/66.34	85.54/74.74	**80.60/67.50**	85.75	87.43	75.42
CMTFNet [6]	89.62/81.19	92.14/85.43	79.88/66.49	85.67/74.93	78.82/65.05	85.23	87.15	74.62
BEDSN [7]	87.51/77.80	90.51/82.65	76.91/62.48	84.23/72.76	66.75/50.09	81.18	85.01	69.16
RS3Mamba [8]	89.52/81.03	92.50/86.04	79.44/65.90	85.62/74.85	78.33/64.38	85.08	87.14	74.44
SAM-FPN-Seg	90.45/82.56	94.14/88.93	79.18/65.53	86.17/75.70	78.96/65.23	85.78	87.97	75.59
HL-SAM-Seg (ours)	**91.21/83.84**	**94.20/89.03**	**80.23/66.99**	**86.21/75.77**	78.46/64.55	**86.06**	**88.37**	**76.04**

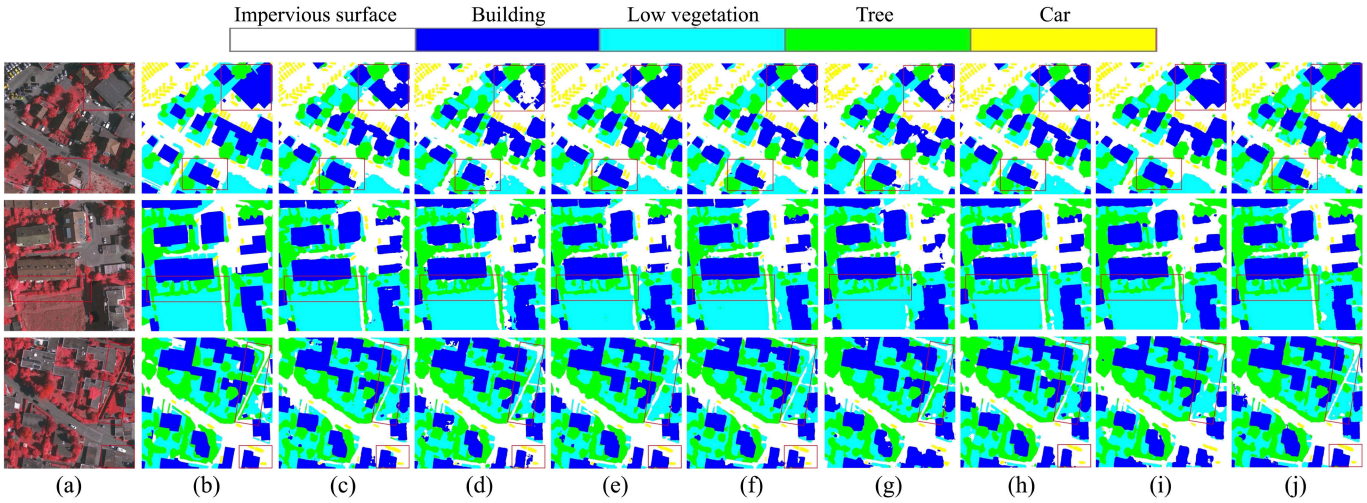


Fig. 6. Segmentation results on the ISPRS Vaihingen dataset with (a) input images, (b) GT, (c) DeepLabv3 +, (d) U-Net, (e) UnetFormer, (f) CMTFNet, (g) BEDSN, (h) RS3Mamba, (i) SAM-FPN-Seg, and (j) HL-SAM-Seg (ours).

IV. RESULTS

A. Comparison With the State-of-the-Art Methods on ISPRS Vaihingen

The experimental results of HL-SAM-Seg on the ISPRS Vaihingen dataset are compared with those of DeepLabv3 +, U-Net, UNetFormer, CMTFNet, BEDSN, RS3Mamba, and RSPrompter in Table I. RSPrompter [24] is a method specifically proposed for remote sensing image segmentation based on the SAM. This model employs a feature pyramid network (FPN)-like structure to extract and aggregate multi-scale features from different hierarchical layers of the SAM backbones and trains a prompter encoder for image segmentation. This architecture functions similar to the image encoder in HL-SAM-Seg by integrating low-level spatial details with high-level semantic information. Since our approach does not change and train the prompt encoder, we replace only the image encoder in Fig. 2 with the FPN-style structure and evaluate its performance compared to our proposed method. We refer to this model as SAM-FPN-Seg in this study, and its results are also included in Table I. In accordance with common practice in related studies [2], [6], the accuracy for the clutter/background category is not reported.

Overall, our proposed HL-SAM-Seg method obtained the highest m$F1$ (86.06%), mIoU (76.04%), and OA (88.37%). Compared with the suboptimal method (SAM-FPN-Seg), m$F1$ is improved by 0.28%, mIoU by 0.45%, and OA by 0.40%. Our method obtained the highest scores in almost all categories. In the impervious surface and low vegetation categories, our method obtained an $F1$ score of 91.21% and 80.23%, respectively, outperforming the second-placed SAM-FPN-Seg by 0.76% and 1.05%.

Fig. 6 presents a comparative visualization of segmentation results from the proposed method and other baseline approaches. As shown, HL-SAM-Seg produces more accurate and structurally coherent segmentation masks, particularly in preserving fine details and maintaining object integrity, most notably for impervious surfaces and buildings. Our method achieves a more complete building segmentation in the first row with clearer boundaries. The second row illustrates that HL-SAM-Seg effectively identifies contiguous tree regions, outperforming other methods in capturing cohesive semantic areas. Finally, the third row demonstrates the model's capability to segment fine-structured roads, a challenging category within impervious surfaces.

B. Comparison With the State-of-the-Art Methods on ISPRS Potsdam

The experimental results of the HL-SAM-Seg on the ISPRS Potsdam dataset are compared with the results derived

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TABLE II

COMPARISON OF RESULTS OF DIFFERENT METHODS ON THE POTSDAM DATASET. IN EACH CATEGORY, THE DIGITS ARE $F1$/MIOU SCORES. THE BEST VALUES IN THE COLUMNS ARE IN BOLD (UNITS: %)

Methods	Impervious surface	Building	Low vegetation	Tree	Car	mF1	OA	mIoU
DeepLabv3+ [1]	91.79/84.83	95.88/92.08	84.72/73.49	84.08/72.53	90.78/83.11	89.45	89.93	81.21
U-Net [2]	92.13/79.83	96.01/86.63	85.48/69.57	84.77/67.48	91.05/80.62	86.71	86.93	76.83
UnetFormer [5]	91.19/83.81	95.75/91.85	83.74/72.03	82.65/70.43	90.47/82.60	88.76	89.26	80.14
CMTFNet [6]	92.13/85.42	96.01/92.31	85.48/74.65	**84.77/73.57**	91.05/83.56	89.89	90.38	81.90
BEDSN [7]	90.57/82.77	94.77/90.07	83.79/72.11	82.07/69.59	90.19/82.14	88.28	88.68	79.33
RS3Mamba [8]	91.30/83.99	95.58/91.53	84.27/72.82	83.18/71.21	90.51/82.66	88.97	89.42	80.44
SAM-FPN-Seg	91.89/84.99	96.11/92.52	85.13/74.12	84.18/72.68	**91.65/84.59**	89.79	90.09	81.78
HL-SAM-Seg (ours)	**92.53/86.10**	**96.44/93.12**	**85.80/75.13**	84.63/73.36	91.22/83.86	**90.12**	**90.65**	**82.31**

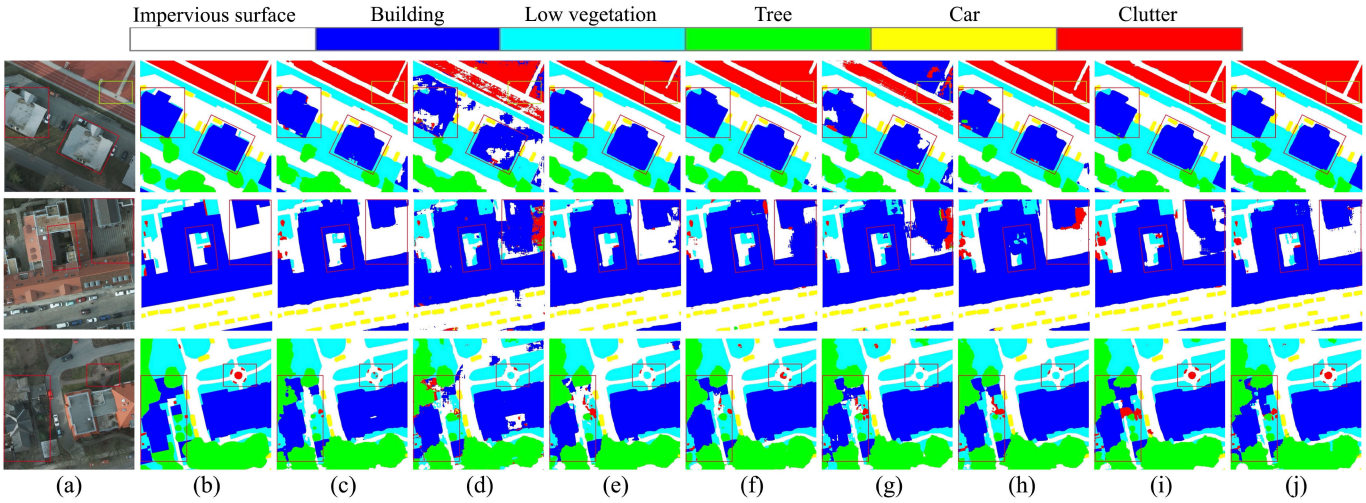


Fig. 7. Segmentation results on the ISPRS Potsdam dataset with (a) input images, (b) GT, (c) DeepLabv3 +, (d) U-Net, (e) UnetFormer, (f) CMTFNet, (g) BEDSN, (h) RS3Mamba, (i) SAM-FPN-Seg, and (j) HL-SAM-Seg (ours).

by DeepLabv3 +, U-Net, UNetFormer, CMTFNet, BEDSN, RS3Mamba, and RSPrompter in Table II. The accuracy of the clutter/background is not reported in the same way as that of the Vaihingen dataset. Overall, our method obtained the highest m$F1$ (90.12%), mIoU (82.31%), and OA (90.65%). Compared to the suboptimal method (CMTFNet), m$F1$ improved by 0.23%, mIoU by 0.41%, and OA by 0.27%. Our method achieved the highest scores in most categories. It achieved an $F1$ score of 92.53% in the impervious surface category and an $F1$ score of 96.44% in the building category.

Also, to visualize the differences between our method and other popular methods, we show three examples of the segmented results in Fig. 7. The proposed HL-SAM-Seg captures more detailed information and can accurately identify the edges of objects compared to other methods. In the first row, our method accurately segments the running lane in the playground and produces more complete building boundaries. In the second row, our method effectively segments the details within the building and exhibits fewer misclassifications in the upper right corner. Specifically, compared to the SAM-FPN-Seg, which is also based on SAM, our approach achieves smoother edge segmentation for the impervious surfaces on the playground. In the third row, our method accurately distinguishes the fine-grained background classes around the roundabout, yielding a building segmentation in the left box closer to the ground truth (GT). Additionally, our method successfully segments the three trees on the right side of the building in the lower left box, while other methods poorly segment the middle tree. Compared to the SAM-FPN-Seg, our approach demonstrates more accurate results in overall detail and edge delineation.

C. Comparison With the State-of-the-Art Methods on FloodNet

The performance of different algorithms on the FloodNet dataset is shown in Table III. Overall, our method achieved the highest scores in m$F1$ (83.55%), mIoU (72.56%), and OA (90.90%). Compared to the suboptimal method, UNetFormer, HL-SAM-Seg improves m$F1$ by 1.39%, mIoU by 1.57%, and OA by 0.26%. Among the different categories, it attains particularly good performance in flooded buildings with the highest $F1$ of 87.93% and trees with the highest $F1$ of 91.03%. It also ranks first for nonflooded buildings, nonflooded roads, water, and grass, demonstrating robustness across almost all categories. However, we observe slightly lower performance

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TABLE III

COMPARISON OF RESULTS OF DIFFERENT METHODS ON THE FLOODNET DATASET. IN EACH CATEGORY, THE DIGITS ARE $F1$/MIOU SCORES. THE BEST VALUES IN THE COLUMNS ARE IN BOLD (UNITS: %)

Methods	Building flooded	Building non-flooded	Road flooded	Road non-flooded	Water	Tree	Vehicle	Pool	Grass	mF1	OA	mIoU
DeepLabv3+ [1]	86.98/ 78.15	86.66/ 76.46	73.48/ 57.61	89.32/ 80.69	77.96/ 63.88	90.30/ 82.32	69.07/ 52.76	**76.70/ 62.21**	93.46/ 87.73	82.66	89.83	71.31
U-Net [2]	79.65/ 66.18	75.00/ 60.01	58.06/ 40.91	83.82/ 72.15	75.13/ 60.17	82.24/ 69.84	53.37/ 36.39	56.37/ 39.24	90.62/ 82.84	72.70	85.31	58.64
UnetFormer [5]	87.01/ 78.20	86.63/ 76.59	**75.30/ 60.34**	89.10/ 80.65	83.87/ 72.22	90.23/ 82.20	62.10/ 44.95	71.69/ 55.88	93.55/ 87.88	82.16	90.64	70.99
CMTFNet [6]	87.53/ 77.83	84.36/ 72.95	63.15/ 46.15	86.65/ 76.44	75.76/ 60.98	88.65/ 79.61	57.25/ 40.11	69.66/ 53.44	90.79/ 83.13	78.20	87.10	65.63
BEDSN [7]	86.10/ 75.59	84.43/ 73.06	33.44/ 73.86	75.69/ 89.30	45.49/ 78.83	76.08/ 90.24	38.84/ 60.40	34.65/ 75.26	84.76/ 92.64	72.77	84.80	59.73
RS3Mamba [8]	87.72/ 78.23	85.68/ 74.95	73.86/ 58.81	89.30/ 80.48	78.83/ 65.06	90.24/ 82.21	60.40/ 43.75	75.26/ 60.34	92.64/ 86.30	81.55	89.29	70.01
SAM-FPN-Seg	87.13/ 77.20	86.60/ 76.36	67.19/ 50.36	88.82/ 79.88	78.68/ 64.50	90.35/ 82.40	**70.60/ 55.23**	76.28/ 61.66	92.80/ 86.57	82.05	89.18	70.46
HL-SAM-Seg (ours)	**87.93/ 78.46**	**86.82/ 76.71**	75.24/ 60.31	**89.41/ 80.85**	**83.93/ 72.31**	**91.03/ 83.55**	67.87/ 51.37	75.96/ 61.24	**93.78/ 88.28**	**83.55**	**90.90**	**72.56**

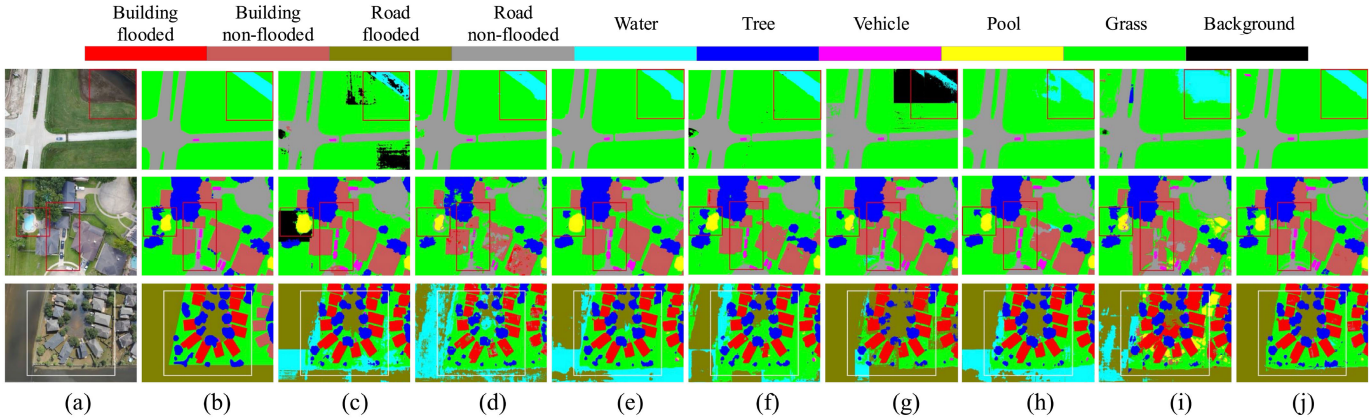


Fig. 8. Segmentation results on the FloodNet dataset with (a) input images, (b) GT, (c) DeepLabv3 +, (d) U-Net, (e) UnetFormer, (f) CMTFNet, (g) BEDSN, (h) RS3Mamba, (i) SAM-FPN-Seg, and (j) HL-SAM-Seg (ours).

in nonflooded buildings compared to UnetFormer. For the vehicle and pool categories, our method performs slightly lower compared to the SAM-FPN-Seg. Similar to the above-mentioned reason, the HL module shows relatively weaker edge perception for relatively independent and discontinuous objects. It is worth noting that an individual pool generally covers a larger area than a vehicle; thus, the performance gap compared to the suboptimal method is less pronounced in this category. Fig. 8 shows three examples of segmentation results derived from these different networks for visualization. It can be observed that the proposed HL-SAM-Seg captures more detailed information and restores more accurate object edges compared to other methods.

Fig. 8 shows three examples of segmentation results derived from these different networks for visualization. It can be observed that the proposed HL-SAM-Seg captures more detailed information and accurately identifies object edges compared to other methods. In the first row, our method segments the water area more accurately and maintains smoother boundaries for the nonflooded roads. In the second row, within the left bounding box, our method achieves relatively precise pool segmentation. In the middle bounding box, it accurately segments both the nonflooded road and the vehicle moving on it. In the third row, which depicts a densely distributed and categorically complex region, our method delivers more accurate segmentation within the bounding box and exhibits fewer misclassifications for flooded roads than other methods.

V. ABLATION STUDY

To evaluate the performance of each module of the proposed HL-SAM-Seg, ablation experiments were conducted on the ISPRS Potsdam dataset. We chose this dataset since it has a more balanced class distribution. The effectiveness of the adapter, OCL, and HL modules was tested by adding them to the baseline one by one, and the experimental results are shown in Table IV.

Baseline: The baseline was constructed on the original SAM model. We only modified the number of MLPs in the mask decoder to generate class masks.

1) *Baseline + Adapter*: The adapter module was introduced into the baseline to adapt the remote sensing semantic segmentation task. As shown in Table V, the baseline

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TABLE IV
ABLATION STUDY OF EACH MODULE OF THE HL-SAM-SEG. IN EACH CATEGORY, THE DIGITS ARE $F1$/MIOU SCORES. THE BEST VALUES IN THE COLUMNS ARE IN BOLD (UNITS: %)

Methods	Impervious surface	Building	Low vegetation	Tree	Car	mF1	OA	mIoU
Baseline	80.79/67.77	88.01/78.59	72.02/56.27	70.45/54.38	84.17/72.66	79.09	78.91	65.94
Baseline+adapter	91.06/83.59	95.02/90.52	83.75/72.05	81.98/69.47	87.42/77.65	87.85	88.79	78.66
Baseline+adapter+OCL	91.16/83.76	95.42/91.24	84.21/72.72	82.14/69.70	89.41/80.85	88.47	89.12	79.65
HL-SAM-Seg (ours)	**92.53/86.10**	**96.44/93.12**	**85.80/75.13**	**84.63/73.36**	**91.22/83.86**	**90.12**	**90.65**	**82.31**

TABLE V
COMPUTATIONAL COMPLEXITY ANALYSIS MEASURED BY A 1024 × 1024 IMAGE

Models	Total Params (M)	Learnable Params (M)	FLOPs (G)	Inference time (FPS)
DeepLabv3+	39.8	39.8	693.68	3.0
U-Net	31	31	876.14	2.8
UNetFormer	**24.2**	24.2	**96.71**	**3.8**
CMTFNet	30.1	30.1	131.07	3.1
BEDSN	17.36	17.36	912.18	**3.8**
RS3Mamba	43.3	43.3	253.20	2.9
SAM-FPN-Seg	104.1	14.4	545.40	1.2
HL-SAM-Seg (ours)	91.7	**5.2**	404.82	1.2

with adapter significantly outperformed the baseline, achieving an m$F1$ of 87.85% (8.76% improvement), an mIoU of 78.66% (12.72% improvement), and an OA of 88.79% (9.88% improvement).

2) *Baseline + Adapter + OCL*: The OCL module was used to generate high-res multiscale features, which were merged in the same way as the operation in Section II-E. As shown in Table V, the OCL module improved the m$F1$ by 0.62%, mIoU by 0.99%, and OA by 0.33%, demonstrating the effectiveness of upsampling and multiscale feature merging.
3) *HL-SAM-Seg*: The OCL and HL modules were introduced to build the entire HL-SAM-Seg network. Table V highlights the significant performance gains achieved by incorporating the HL module. It led to a 1.32% increase in m$F1$, a 2.12% increase in mIoU, and a 1.23% increase in OA, underscoring its effectiveness. The HL-SAM-Seg network outperformed all other models across all categories.

Fig. 9 shows three examples of the ablation results for each module. A comparison between Fig. 9(b) and (c) reveals that the network had a high probability of misclassification when the adapter module was not added. For example, in the three examples, large parts of the low vegetation and impervious surface are misclassified as buildings in Fig. 9(b). In the first row, the upper right part of the playground is misclassified as low vegetation in Fig. 9(b). Comparing Fig. 9(c) and (d), the OCL module improves detail performance due to high-res feature incorporation. However, due to a lack of low-res semantic association, it can misclassify regions, such as the playground in the first row, column (c). The HL module addresses this issue. It can be seen from the comparison between Fig. 9(b) and (f) that the network underperforms in segmenting fine details when the HL module is not added. For instance, in the first row, the building and playground are more finely segmented in Fig. 9(f) than in the other columns. In the second row, the low vegetation within the building area is segmented in finer detail, and the roundabout in the third row is also accurately delineated. Additionally, the association maps between the original-size high-res features and low-res features are presented in Fig. 9(e). It can be observed from Fig. 9(e) that high-res pixels are assigned to their corresponding semantic objects, indicating that the high-res features are effectively associated with low-res semantics.

VI. DISCUSSION

A. Model Complexities

The above experiments are all carried out on a single NVIDIA GeForce RTX 3090 GPU. Table V shows the computational complexity measured by a 1024 × 1024 image. For the HL-SAM-Seg network, although the total number of parameters reaches 91.7 million, much more than the UNetFormer, the number of its learnable parameters is only 5.2 million, the least among all the compared methods. It should be noted that the high floating-point operations (FLOPs) of the HL-SAM-Seg are due to the high computational cost of the SAM backbone. The FLOPs are fewer than those of the same SAM-based network, SAM-FPN-Seg. The inference time is calculated to be sample pictures per second, which can be regarded as frames per second (FPS). The proposed HL-SAM-Seg is not the fastest way, which is also due to the high computational cost of the SAM backbone.

B. Effectiveness of the HL Module

In this study, we leverage the pretrained SAM backbone to extract features for remote sensing semantic segmentation. In addition to the original transformer modules in the SAM feature encoder, a high-res path is added to obtain high-res features. To integrate the high-res and low-res features, we design an HL module encompassing the cross-attention mechanism to construct and update the relationship between high-res and low-res features. This is similar to the trainable superpixel method [38], [39], which maintains an association map between pixels and superpixels.

To explicitly visualize the effects of the HL module on the extracted features, we compare the feature maps before and after passing through the three HL modules, as demonstrated

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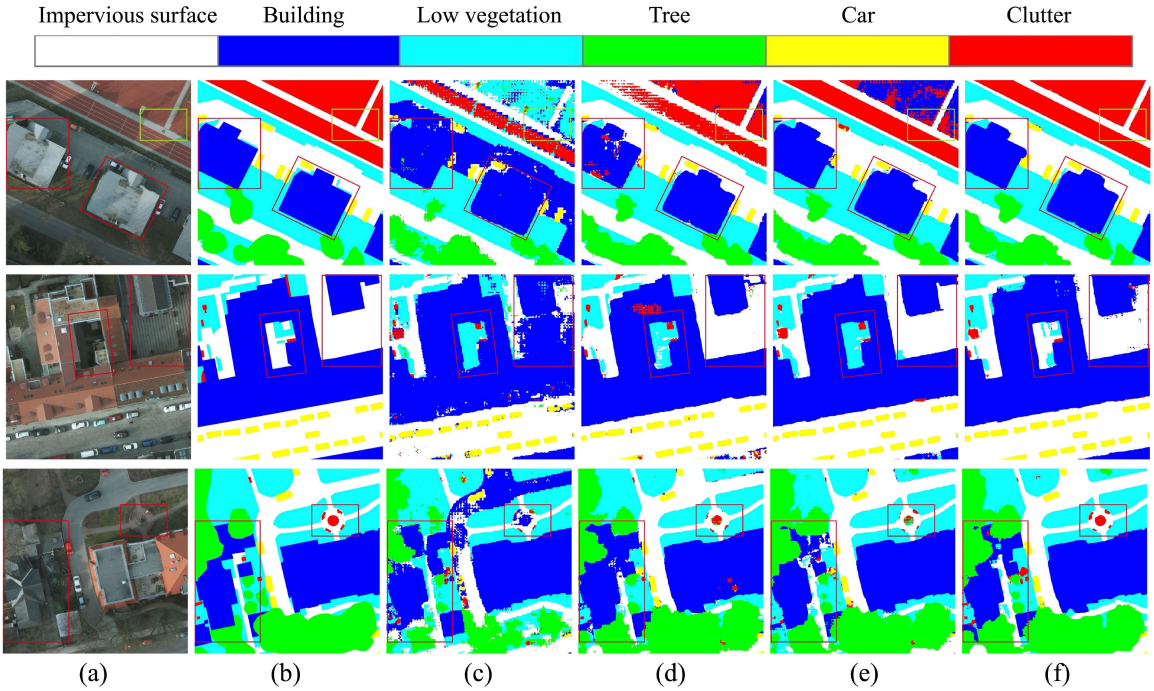


Fig. 9. Ablation experimental results on the ISPRS Potsdam dataset. (a) Input images. (b) GT. (c) SAM baseline. (d) SAM baseline with adapter. (e) SAM baseline with adapter and OCL. (f) HL-SAM-Seg network.

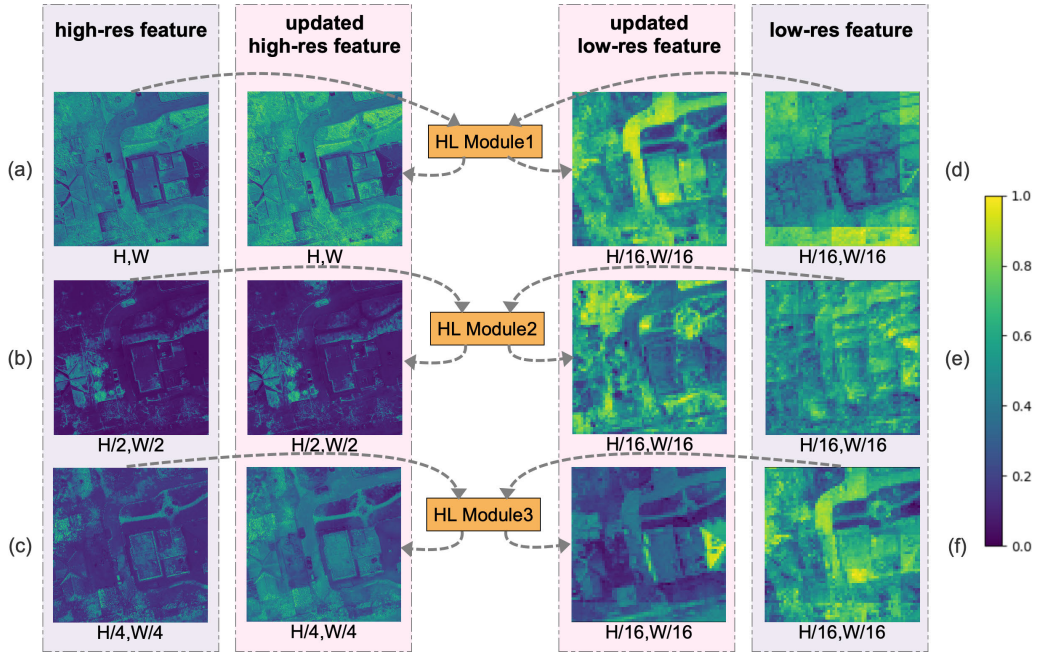


Fig. 10. Visualization of the normalized feature map before and after passing through the HL module. (a)–(c) High-res features at scales 1, 1/2, and 1/4 of the input image size. (b)–(f) Low-res features at scale 1/16 of the input image size.

in Fig. 10. It can be seen from the left two columns in Fig. 10 that the high-res features updated by the HL module exhibit more consistent and continuous feature values. From Fig. 10(a) and (b), we can see that the HL modules 1 and 2 only slightly change the feature values in the updated high-res feature. This effect gradually accumulates and becomes more pronounced after the feature map goes through the HL module 3, as seen in Fig. 10(c), where the buildings become more consistent, and the difference between different semantic objects becomes more distinct. For the updating of low-res features, as shown in the right two columns of Fig. 10, the improvements of the updated low-res features are more obvious. After each HL module, the updated low-res features align more closely with the actual targets in terms of both semantics and contours.

C. Limitations in Small Sample Categories

From the results listed in Tables I–III, we can observe the underperformance of the HL-SAM-Seg in trees and cars of the ISPRS datasets, as well as flooded roads, vehicles, and pools of the FloodNet dataset. This is primarily due to

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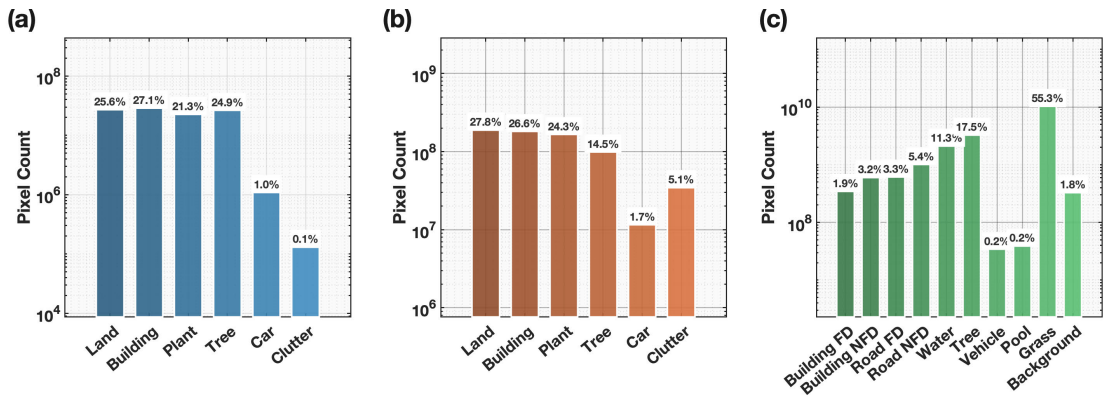


Fig. 11. Class distribution of the training datasets for (a) ISPRS Vaihingen, (b) ISPRS Potsdam, and (c) FloodNet datasets (FD means flooded and NFD denotes nonflooded).

their small sample sizes in the training datasets, as shown in the class distributions in Fig. 11. Although we use data augmentation in this study, it is not aimed at balancing the samples of different categories. Possible solutions will be tested in future work to address this small sample problem, including data augmentation to expand the number of small samples, generative adversarial networks (GANs) that learn to generate new samples, or class-balanced loss functions, potentially incorporating focal loss mechanisms or multiscale loss functions that emphasize the learning of small-scale features.

VII. Conclusion

In this study, we propose the HL-SAM-Seg, a remote sensing image semantic segmentation model based on SAM that combines high-res and low-res features. This model leverages the powerful visual prior knowledge encoded in the pretrained SAM backbone and efficiently fuses multiresolution features through the proposed HL module. We conducted comparative and ablation experiments on three high-resolution remote sensing datasets, ISPRS Vaihingen, ISPRS Potsdam, and FloodNet, demonstrating the effectiveness of our proposed method by comparing it against the other six state-of-the-art semantic segmentation models in the overall m$F1$, mIoU, and OA. Specifically, it achieves improvements of 0.28%, 0.45%, and 0.40% in m$F1$, mIoU, and OA on the ISPRS Vaihingen dataset; 0.23%, 0.41%, and 0.27% on the ISPRS Potsdam dataset; and 1.39%, 1.57%, and 0.26% on the FloodNet dataset. The segmented results more accurately preserve fine details and contours of semantic objects, such as buildings, impervious surfaces, and low vegetation, than the competing methods. Furthermore, the proposed model can be trained efficiently with only a small number of learnable parameters by keeping the pretrained SAM backbone frozen, underscoring the practicality of leveraging large pretrained models for downstream tasks in remote sensing. Due to its robustness and high accuracy, HL-SAM-Seg holds strong potential for various remote sensing applications, such as urban change detection, road extraction, and building segmentation, which will be explored in future work.

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